

India generates approximately 500 million tonnes of agricultural waste annually. A substantial portion of this waste, particularly in states like Punjab, Haryana, and Uttar Pradesh, is burned in open fields due to the lack of economically viable and logistically feasible alternatives. This practice results in:

- **Air Pollution:** Stubble burning contributes to severe air quality deterioration, with PM 2.5 levels reaching hazardous levels in the Indo-Gangetic plains.
- **Health Hazards:** Respiratory diseases, cardiovascular issues, and allergic reactions increase significantly during the burning season.
- **Soil Degradation:** Burning destroys beneficial soil microorganisms, reduces organic carbon content, and diminishes soil fertility over time.
- **Greenhouse Gas Emissions:** Agricultural burning releases CO₂, CO, CH₄, and N₂O, contributing to climate change.

2.1 State-wise Crop Residue Generation

The following table presents state-wise crop residue generation and estimated residue burned, based on data from the Indian Agricultural Research Institute (IARI) and the Ministry of Agriculture:

Table 2.1: State-wise Crop Residue Generation and Burning Estimates

State	Residue Generated (MT)	Residue Burned (MT)	% Burned
Punjab	51.0	19.8	38.8
Haryana	27.6	9.1	33.0
Uttar Pradesh	59.0	21.3	36.1
Maharashtra	46.5	7.4	15.9
Madhya Pradesh	33.2	4.3	12.9
Karnataka	28.4	3.8	13.4
Andhra Pradesh	25.1	2.5	10.0
Tamil Nadu	19.8	3.9	19.7
Others	209.4	20.0	9.5
Total	500.0	92.1	18.4

2.2 Limitations of Existing Solutions

Several initiatives have been undertaken to address agricultural waste management, but they face significant limitations: